

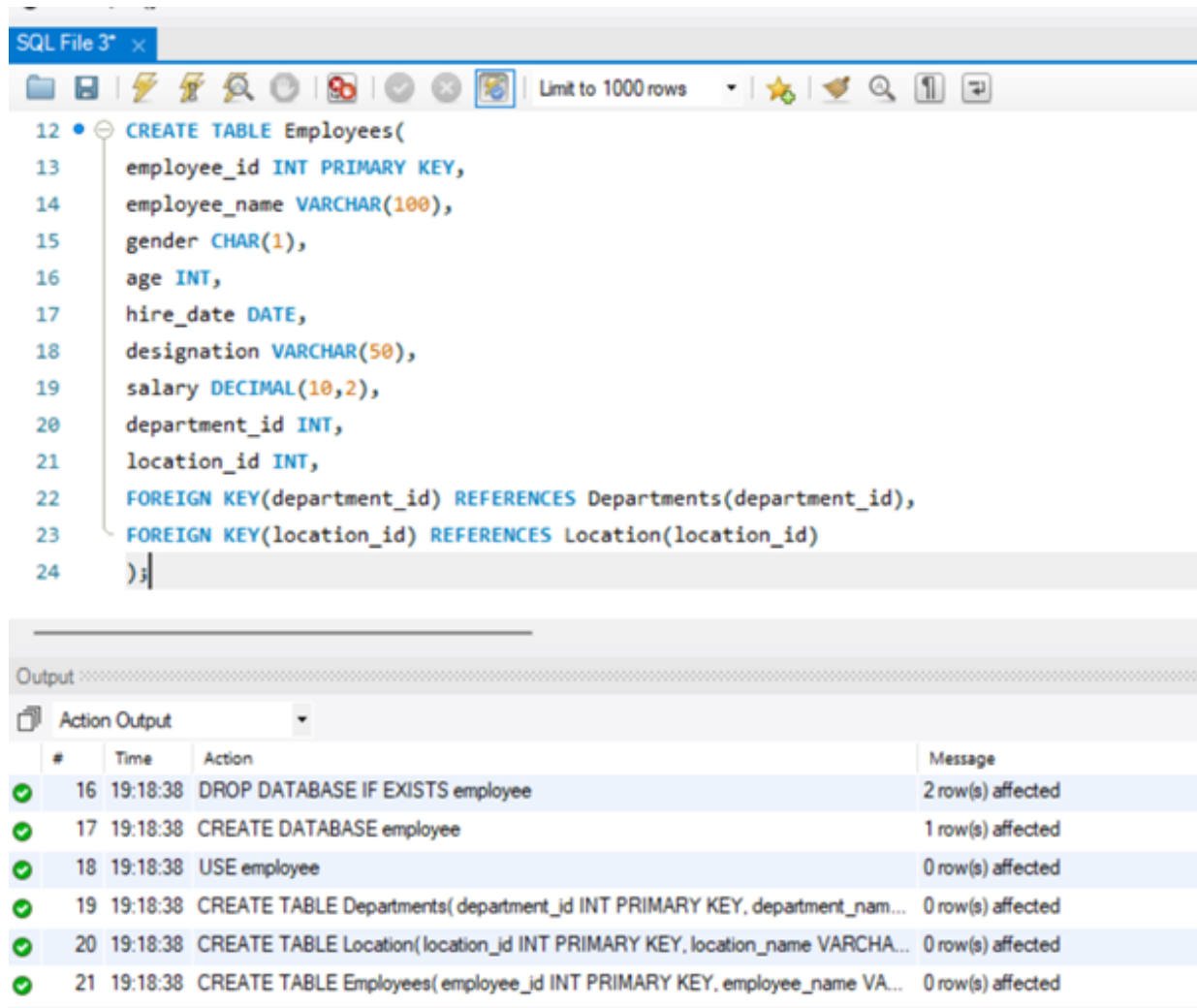
SQL ASSIGNMENT REPORT

DDL Commands and Constraints in MySQL

Objective

To create a database, create tables, perform ALTER/RENAME/TRUNCATE/DROP operations, and implement constraints in MySQL.

DDL - Create Database and Departments Table



The screenshot displays a MySQL IDE interface. The top pane shows the following SQL code:

```
12 CREATE TABLE Employees(  
13     employee_id INT PRIMARY KEY,  
14     employee_name VARCHAR(100),  
15     gender CHAR(1),  
16     age INT,  
17     hire_date DATE,  
18     designation VARCHAR(50),  
19     salary DECIMAL(10,2),  
20     department_id INT,  
21     location_id INT,  
22     FOREIGN KEY(department_id) REFERENCES Departments(department_id),  
23     FOREIGN KEY(location_id) REFERENCES Location(location_id)  
24 );
```

The bottom pane shows the 'Output' window with the 'Action Output' tab selected. It contains a table with the following data:

#	Time	Action	Message
✓ 16	19:18:38	DROP DATABASE IF EXISTS employee	2 row(s) affected
✓ 17	19:18:38	CREATE DATABASE employee	1 row(s) affected
✓ 18	19:18:38	USE employee	0 row(s) affected
✓ 19	19:18:38	CREATE TABLE Departments(department_id INT PRIMARY KEY, department_nam...	0 row(s) affected
✓ 20	19:18:38	CREATE TABLE Location(location_id INT PRIMARY KEY, location_name VARCHA...	0 row(s) affected
✓ 21	19:18:38	CREATE TABLE Employees(employee_id INT PRIMARY KEY, employee_name VA...	0 row(s) affected

DDL - Create Employees Table

```
SQL File 3* x
Limit to 1000 rows
23 FOREIGN KEY(location_id) REFERENCES Location(location_id)
24 );
25 • ALTER TABLE Employees
26 ADD email VARCHAR(100);
27 • ALTER TABLE Employees
28 MODIFY designation VARCHAR(100);
29 • ALTER TABLE Employees
30 DROP COLUMN age;
```

DDL - ALTER and RENAME Commands

- ALTER TABLE Employees
RENAME COLUMN hire_date TO date_of_joining;
- RENAME TABLE Departments TO Departments_Info;
- RENAME TABLE Location TO Locations;

Action Output

	Time	Action	Message
73	19:26:41	ALTER TABLE Employees ADD email VARCHAR(100)	0 row(s)
74	19:26:41	ALTER TABLE Employees MODIFY designation VARCHAR(100)	0 row(s)
75	19:26:41	ALTER TABLE Employees DROP COLUMN age	0 row(s)
76	19:26:41	ALTER TABLE Employees RENAME COLUMN hire_date TO date_of_joining	0 row(s)
77	19:26:41	RENAME TABLE Departments TO Departments_Info	0 row(s)
78	19:26:41	RENAME TABLE Location TO Locations	0 row(s)

DDL - TRUNCATE and DROP Commands

The screenshot shows a SQL script in a file named 'SQL File 3*'. The script contains the following commands:

```
24 );
25 ALTER TABLE Employees
26 ADD email VARCHAR(100);
27 ALTER TABLE Employees
28 MODIFY designation VARCHAR(100);
29 ALTER TABLE Employees
30 DROP COLUMN age;
31 ALTER TABLE Employees
32 RENAME COLUMN hire_date TO date_of_joining;
33 RENAME TABLE Departments TO Departments_Info;
34 RENAME TABLE Location TO Locations;
35 TRUNCATE TABLE Employees;
36 DROP TABLE Employees;
37
```

The Output window shows the execution results:

#	Time	Action	Message
100	19:34:03	ALTER TABLE Employees DROP COLUMN age	0 row
101	19:34:03	ALTER TABLE Employees RENAME COLUMN hire_date TO date_of_joining	0 row
102	19:34:03	RENAME TABLE Departments TO Departments_Info	0 row
103	19:34:03	RENAME TABLE Location TO Locations	0 row
104	19:34:03	TRUNCATE TABLE Employees	0 row
105	19:34:03	DROP TABLE Employees	0 row

Constraints - Create Tables

The screenshot shows a SQL script in a file named 'SQL File 3*'. The script contains the following command:

```
49 CREATE TABLE Employees(
50     employee_id INT PRIMARY KEY,
51     employee_name VARCHAR(100) NOT NULL,
52     gender CHAR(1),
53     age INT CHECK(age>=18),
54     hire_date DATE DEFAULT(CURRENT_DATE),
55     designation VARCHAR(100),
56     salary DECIMAL(10,2),
57     department_id INT,
58     location_id INT,
59     CHECK(gender IN('M','F')),
60     FOREIGN KEY(department_id) REFERENCES Departments(department_id),
61     FOREIGN KEY(location_id) REFERENCES Locations(location_id)
62 );
```

The Output window shows the execution results:

#	Time	Action	Message
226	19:40:34	DROP DATABASE IF EXISTS employee	0 row(s) affected, 1 warning(s): 1008
227	19:40:34	CREATE DATABASE employee	1 row(s) affected
228	19:40:34	USE employee	0 row(s) affected
229	19:40:34	CREATE TABLE Departments(department_id INT PRIMARY KEY, department_name VAR...	0 row(s) affected
230	19:40:34	CREATE TABLE Locations(location_id INT AUTO_INCREMENT PRIMARY KEY, location...	0 row(s) affected
231	19:40:34	CREATE TABLE Employees(employee_id INT PRIMARY KEY, employee_name VARCHA...	0 row(s) affected

SHOW TABLES Output

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

Tables_in_employee

- departments
- employees
- locations

Result 1 x

Output

Action Output

#	Time	Action	Message
✓ 248	19:44:49	CREATE DATABASE employee	1 row(s) affected
✓ 249	19:44:49	USE employee	0 row(s) affected
✓ 250	19:44:49	CREATE TABLE Departments(department_id INT PRIMARY KEY, department_name VA...	0 row(s) affected
✓ 251	19:44:49	CREATE TABLE Locations(location_id INT AUTO_INCREMENT PRIMARY KEY, locatio...	0 row(s) affected
✓ 252	19:44:49	CREATE TABLE Employees(employee_id INT PRIMARY KEY, employee_name VARCHA...	0 row(s) affected
✓ 253	19:44:49	SHOW TABLES	3 row(s) returned

DESC Departments

```
59 CHECK(gender IN('M','F'))),
60 FOREIGN KEY(department_id) REFERENCES Departments(department_id),
61 FOREIGN KEY(location_id) REFERENCES Locations(location_id)
62 );
63 SHOW TABLES;
64 DESC Departments;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

Field	Type	Null	Key	Default	Extra
department_id	int	NO	PRI		
department_name	varchar(100)	NO	UNI		

Result 2 Result 3 x

Output

Action Output

#	Time	Action	Message
✓ 271	19:46:25	USE employee	0 row(s) affected
✓ 272	19:46:25	CREATE TABLE Departments(department_id INT PRIMARY KEY, department_name VA...	0 row(s) affected
✓ 273	19:46:25	CREATE TABLE Locations(location_id INT AUTO_INCREMENT PRIMARY KEY, locatio...	0 row(s) affected
✓ 274	19:46:25	CREATE TABLE Employees(employee_id INT PRIMARY KEY, employee_name VARCHA...	0 row(s) affected
✓ 275	19:46:25	SHOW TABLES	3 row(s) returned
✓ 276	19:46:25	DESC Departments	2 row(s) returned

DESC Locations

```
61 FOREIGN KEY(location_id) REFERENCES Locations(location_id)
62 );
63 • SHOW TABLES;
64 • DESC Departments;
65 • DESC Locations;
```

Field	Type	Null	Key	Default	Extra
location_id	int	NO	PRI	HULL	auto_increment
location_name	varchar(100)	NO	UNI	HULL	



Result 4 Result 5 Result 6 x

Output

📄 Action Output ▾

	#	Time	Action	Message
✓	295	19:47:20	CREATE TABLE Departments(department_id INT PRIMARY KEY, department_name VA...	0 row(s) affected
✓	296	19:47:20	CREATE TABLE Locations(location_id INT AUTO_INCREMENT PRIMARY KEY, locatio...	0 row(s) affected
✓	297	19:47:20	CREATE TABLE Employees(employee_id INT PRIMARY KEY, employee_name VARCHA...	0 row(s) affected
✓	298	19:47:20	SHOW TABLES	3 row(s) returned
✓	299	19:47:20	DESC Departments	2 row(s) returned
✓	300	19:47:20	DESC Locations	2 row(s) returned

DESC Employees

The screenshot shows a MySQL IDE window with a SQL script and its execution results. The script includes a foreign key constraint and several descriptive commands. The results pane shows the output of these commands, including the structure of the 'Employees' table.

```
61 FOREIGN KEY(location_id) REFERENCES Locations(location_id)
62 );
63 • SHOW TABLES;
64 • DESC Departments;
65 • DESC Locations;
66 • DESC Employees;
67
```

Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	<u>NULL</u>	
employee_name	varchar(100)	NO		<u>NULL</u>	
gender	char(1)	YES		<u>NULL</u>	
age	int	YES		<u>NULL</u>	
hire_date	date	YES		<u>curdate()</u>	DEFAULT_GENERATED

#	Time	Action	Message
✓ 320	19:48:13	CREATE TABLE Locations(location_id INT AUTO_INCREMENT PRIMARY KEY, locatio...	0 row(s) affected
✓ 321	19:48:13	CREATE TABLE Employees(employee_id INT PRIMARY KEY, employee_name VARCHA...	0 row(s) affected
✓ 322	19:48:13	SHOW TABLES	3 row(s) returned
✓ 323	19:48:13	DESC Departments	2 row(s) returned
✓ 324	19:48:13	DESC Locations	2 row(s) returned
✓ 325	19:48:13	DESC Employees	9 row(s) returned

Conclusion

The Employee database was successfully created using MySQL. DDL operations and required constraints were implemented successfully. The database structure was verified using SHOW TABLES and DESC commands.